

Type Conversions – Simple Hinglish Summary

Source: <https://javascript.info/type-conversions>

Main baat

Operators aur functions zyadatar values ko **automatically sahi type me convert** kar dete hain. Jaise `alert` har value ko string me badalkar dikhata hai, aur math operations values ko number me badalte hain. Kabhi kabhi hume **khud convert** karna padta hai.

(Is chapter me sirf **primitives** ki baat hai, objects baad me.)

1. String Conversion

Jab value ka string form chahiye (jaise `alert` me), tab hota hai. Khud karne ke liye `String(value)`:

```
javascript

let value = true;
alert(typeof value); // boolean

value = String(value); // ab "true" (string)
alert(typeof value); // string
```

Zyadatar obvious hai: `false` → `"false"`, `null` → `"null"`.

2. Numeric Conversion

Math operations me **automatically** hota hai:

```
javascript

alert( "6" / "2" ); // 3
```

Khud karne ke liye `Number(value)`:

```
javascript

let str = "123";
let num = Number(str); // 123 (number)
```

Ye tab zaruri hota hai jab form/text input se value string me aati hai lekin number chahiye.

Agar string valid number nahi hai to result `NaN` aata hai:

javascript

```
let age = Number("an arbitrary string instead of a number");
alert(age); // NaN
```

Numeric conversion ke rules

Value	Kya banti hai
undefined	NaN
null	0
true / false	1 / 0
string	Start aur end ki whitespace hat jati hai. Bacha string khali ho to 0, warna number "padha" jata hai. Error par NaN.

javascript

```
alert( Number(" 123 ") ); // 123
alert( Number("123z") ); // NaN
alert( Number(true) ); // 1
alert( Number(false) ); // 0
```

Dhyan: `null` → `0` lekin `undefined` → `NaN`. Dono alag behave karte hain.

3. Boolean Conversion

Sabse simple. Logical operations me automatically hota hai, khud karne ke liye

`Boolean(value)`.

Rule:

- "Khali" values `false` ban jati hain: `0`, `""` (empty string), `null`, `undefined`, `NaN`
- Baaki sab `true`

javascript

```
alert( Boolean(1) ); // true
alert( Boolean(0) ); // false
alert( Boolean("hello") ); // true
alert( Boolean("") ); // false
```

Trap: `"0"` aur `" "` (sirf space) dono `true` hain. JavaScript me koi bhi non-empty string `true` hoti hai (PHP me `"0"` false hota hai, JS me nahi).

```
javascript

alert( Boolean("0") ); // true
alert( Boolean(" ") ); // true
```

Summary

Conversion	Kab hota hai	Khud kaise karein
String	Output karte waqt	<code>String(value)</code>
Number	Math operations me	<code>Number(value)</code>
Boolean	Logical operations me	<code>Boolean(value)</code>

Boolean rules

Value	Result
<code>0</code> , <code>null</code> , <code>undefined</code> , <code>NaN</code> , <code>""</code>	<code>false</code>
Baaki sab	<code>true</code>

Jahan log galti karte hain

- `undefined` number me `NaN` banta hai, `0` nahi.
- `"0"` aur `" "` boolean me `true` hote hain.